Pivot experiment report

Executive Summary

The clearest finding is that **tone affects both model behavior and performance**. Neutral prompts performed best overall, while polite and aggressive tones produced longer, slower, more explanatory outputs and lower average test performance.

`Role_framing` and `Format` had much smaller effects. The evidence does not strongly support that adding a developer role improves correctness. Plaintext was slightly better than JSON/XML on average, but format effects were modest and less consistent across models.

Data Analyzed

- Result files: `results/gemma2-9b_final.jsonl`, `results/llama3.1-8b_final.jsonl`, `results/qwen2.5-7b_final.jsonl`
- Models: 3
- Problems: 50
- Prompt templates: 18
- Total runs: 2,700
- Prompt factors analyzed: `Tone`, `Role_framing`, `Format`
- `Pressure` was not analyzed because it was empty/constant in the prompt matrix.

Overall Model Performance

Model	Runs	Pass rate	Avg. tests passed	Avg. duration	Avg. completion tokens	Code-only rate
qwen2.5-7b	900	13.4%	17.0%	18.59s	653	26.3%
gemma2-9b	900	9.4%	14.0%	19.56s	601	25.9%
llama3.1-8b	900	3.9%	6.4%	16.33s	562	12.2%

The absolute pass rates are low, so the more sensitive metric is often `tests_fraction`, the fraction of unit tests passed. Exact pass/fail is still useful, but it loses information when many outputs fail at least one test.

Main Finding: Neutral Tone Performs Best

Across all models and templates, neutral prompts had the strongest performance.

Tone	Runs	Pass rate	Avg. tests passed	Avg. duration	Avg. completion tokens	Code-only rate
neutral	900	10.9%	15.4%	12.62s	413	62.3%
polite	900	8.0%	11.2%	21.47s	716	1.4%
aggressive	900	7.9%	10.8%	20.40s	687	0.7%

The paired comparison, which compares prompt levels within the same `(model, problem)` pairs, shows:

Comparison	Pass-rate delta	Tests-fraction delta
polite vs neutral	-2.9 percentage points	-4.2 percentage points
aggressive vs neutral	-3.0 percentage points	-4.6 percentage points

The fixed-effect regression supports the same conclusion. With aggressive tone as the hidden reference category, neutral tone was associated with:

- `+3.0` percentage points on exact pass rate (`p = 0.0123`)
- `+4.6` percentage points on tests passed fraction (`p = 0.0003`)

Polite tone was not meaningfully different from aggressive tone on correctness in the regression.

Tone Also Changes Behavior

Tone had a much larger behavioral effect than role framing or output format.

Compared with neutral prompts, polite and aggressive prompts generated much longer outputs:

Comparison	Completion-token delta	Duration delta	Code-only-rate delta
polite vs neutral	+304 tokens	+8.85s	-60.9 percentage points
aggressive vs neutral	+274 tokens	+7.78s	-61.7 percentage points

This suggests that polite/aggressive prompt variants encouraged more prose, explanation, or wrapper text. That behavior likely hurt the coding benchmark because the task expects executable code, and extra text can interfere with extraction or evaluation.

Role Framing Has Little Evidence of Benefit

Role framing had a small effect overall.

Role framing	Runs	Pass rate	Avg. tests passed	Avg. completion tokens	Code-only rate
no role	1,350	9.3%	12.7%	605	22.2%
developer role	1,350	8.5%	12.2%	605	20.7%

The paired comparison found developer role was slightly worse than no role:

- exact pass rate: `-0.8` percentage points
- tests fraction: `-0.5` percentage points

These differences are small, and the regression p-values were not statistically convincing. The practical interpretation is: **developer role framing did not clearly improve performance in this experiment**.

Format Effects Are Modest and Model-Specific

Plaintext was slightly stronger overall than JSON/XML, but the effect was smaller than tone.

Format	Runs	Pass rate	Avg. tests passed	Avg. completion tokens	Code-only rate
plaintext	900	9.4%	13.3%	603	21.0%
xml	900	9.1%	12.4%	601	24.3%
json	900	8.2%	11.7%	612	19.1%

Paired effects versus plaintext:

Comparison	Pass-rate delta	Tests-fraction delta
JSON vs plaintext	-1.2 percentage points	-1.6 percentage points
XML vs plaintext	-0.3 percentage points	-0.9 percentage points

These are modest and their confidence intervals overlap zero. Format may still matter for individual models: for example, qwen2.5-7b did much better with plaintext than JSON/XML, while gemma2-9b showed stronger average tests fraction with XML.

Best and Worst Prompt Templates

The best complete template was:

Template	Tone	Role	Format	Pass rate	Avg. tests passed
1	neutral	no role	plaintext	13.3%	18.4%

Other strong templates were also mostly neutral:

Template	Tone	Role	Format	Pass rate	Avg. tests passed
5	neutral	developer role	json	12.7%	17.7%
6	neutral	developer role	xml	12.7%	16.1%
3	neutral	no role	xml	11.3%	15.2%

The weakest templates were mostly polite or aggressive, especially with developer role or JSON/XML formatting.

Interpretation

The main behavioral mechanism appears to be verbosity. Neutral prompts produced shorter, more code-focused answers and had the highest code-only rate. Polite and aggressive prompts produced hundreds more completion tokens on average, took longer, and were much less likely to return code-only outputs.

For this benchmark, the safest prompt technique is:

> neutral tone + simple/plaintext instructions + no unnecessary persona or role pressure

That combination best matches the evaluator's needs: concise executable code with minimal extra prose.